

Image Classification [Code Challenge MCQ] (Version : 1)

TEST

● **Correct Answer**

🕒 Answered in 1.35 Minutes

Question 1/10

After extracting a subset of the MNIST dataset, what are the shapes of X_train and X_test respectively?

☐ (5000, 28, 28), (1000, 28, 28)

☐ (28, 28, 5000), (28, 28, 1000)

☒ (5000, 784), (1000, 784)

☐ (5000,), (1000,)

Question 2/10

After extracting the data in Question 1, print out the value of X_train[1, 349]. What is this value?

☒ 0.10980392

☐ 0.0

☐ 0.7764706

☐ 0.84313726

Question 3/10

After extracting the data in Question 1, print out the label in `y_test[50]`. What is this label?

☐ 1

☐ 2

☐ 8

☒ 6

Question 4/10

What is the purpose of dividing the image pixel values by 255 in the `get_data` function?

☐ To increase the image size

☐ To reduce the computational complexity without having to normalise the data

☒ To normalise the pixel values between 0 and 1

☐ To convert the images into grayscale

Question 5/10

What is the accuracy calculated in Question 3?

☐ 0.84

☐ 0.09

☐ 0.8

☒ 0.9

Question 6/10

What is the precision calculated for label 6 in Question 3?

☐ 0.98

☐ 0.86

☐ 0.88

☒ 0.93

Question 7/10

What is the f1-score for label 0 calculated in Question 3?

☐ 0.84

☐ 0.92

☒ 0.96

☐ 0.98

Question 8/10

Based on the classification report generated, how many observations in the test set were labeled as '7'?

☐ 1000

☒ 99

☐ 0.89

☐ 85

Question 9/10

What is the predicted label for image index 60?

☐ 5

☐ 9

7

☒

☐ 3

Question 10/10

Which of the following is not a possible outcome of increasing the number of training images used to train the random forest model from 5,000 to 20,000?

☒ A decrease in the amount of time required to train the model due to more efficient learning.

☐ A potential increase in the model's training time because of the larger amount of data to be processed.

☐

A slightly more complex model may be needed to capture the details and variations in the larger dataset.

☐

The model's accuracy on the test set could potentially improve because of exposure to a wider set of examples.